Clemens Scherzer

https://www.linkedin.com/in/clemens-scherzer-73954b2a8/
https://clemensj.github.io (617) 922-0150

Education

Yale University *2024 - Present*
Candidate for BS in Mechanical Engineering (ABET), Varsity Athlete (Crew) GPA: 4.0
Noble and Greenough School *2018 - 2024*
Highest Distinction GPA: 11.0/11.0

Skills

Design & Analysis: NX, Teamcenter, SolidWorks, 3DX, OnShape, GD&T, ANSYS Workbench, Mechanical, Fluent, CFX
Programing: Python, Java, VBA, Excel API, SolidWorks API, MATLAB, Arduino, Ardupilot, Px4
Manufacturing: Lathe, Mill, Shop machines, Power tools, Soldering, FDM, SLA, SLM, MJF printing

Work Experience

SPACEX | Starbase, TX *June 2026 - September 2026*
Booster Vehicle Design Intern

- Worked across analysis and manufacturing teams to upgrade booster capability and reliability.
- Designed upgraded capability sheet metal joiners and stringers for gross acreage, fuel tank door, and grid fins, working with analysis teams to reduce shear, buckling, and thermally induced failure modes.
- Coordinated with propulsion and structures teams to integrate raceway DFM and capability upgrades across all tank stations, including ownership of high level assembly releases.

DRAPER | Boston, MA *June 2025 - August 2025*
Mechanism and Electromechanical Design Intern

- Worked across teams to support electronics development and testing for defense applications.
- Automated creation of precision cut and form tooling for quick-turn IC development using parametric modeling and Solidworks API, resulting in auto-generated models and GD&T drawings.
- Developed cooling system data infrastructure and control software for Celestial Navigation System (CNS) using TCP and Modbus.
- Designed and analyzed mass model of CNS system to meet prime contractor requirements, performed fatigue, shock response spectrum, modal, and functional shock analysis using ANSYS and SolidWorks.

UNITZERO | New Haven, CT *December 2025 - June 2026*
Robotics Consultant

- Designed Universal Manipulator Interface (UMI) with Camera, VR tracker, and Jetson Nano for VLA training.
- Designed 3D-printed 6-DOF hall-effect based leader arm for robot arm control and data collection.

Project Experience

YALE UNDERGRADUATE AEROSPACE ASSOCIATION | New Haven, CT
Project Lead, Yale Jet Aeronautics *January 2025 - Present*

- Attempting to break the sound barrier with an air-breathing vehicle.
- Leading the design, manufacturing, and testing of 350N afterburning jet engine and airframe.
- Performed Turbomachinery design and optimization using CFX.

Aerostructures Lead, Project Liquid *September 2024 - June 2026*

- Responsible for airframe design, packaging, and regeneratively cooled liquid rocket, subteam leadership.

FIRST ROBOTICS COMPETITION TEAM 125 - NUTRONS | Revere, MA *December 2017 - June 2024*
Team Captain, Team Leader of Design and Systems Integration, Subsystem Design Lead

- Led team to finals of world championship, 12 event wins.
- Owned overall robot integration, coordination between hardware, software, and manufacturing subteams.
- Designed launchers, feeders, drivetrains, telescoping arms. Performed handcalcs, FEA, led CAD.

PERSONAL PROJECTS
Hybrid VTOL *August 2023 - December 2024*

- Designed and built patent-pending wingeron/tricopter hybrid capable of plane-like forward flight, efficient hover, and decoupled pitch & translation. Successfully evaluated forward and tricopter flight modes.
- Wrote custom flightcontrol software for Teensy microcontroller.

High-Efficiency FPV Racing Wing *October 2023 - January 2024*

- Designed, optimized, and built FPV racing wing capable of 102mph flight on only 3s LiPo powertrain.
- Used Xflr5 for airfoil, sweep, and wash optimization, longitudinal stability analysis.

Extracurricular Activities and Awards

- National Merit Scholarship Finalist *2024*
- First Robotics Einstein World Championship Finalists *2023*
- Team Captain, Noble and Greenough Boys Varsity Crew *2024*